Beyond Binary Emotions: A Multi-Architecture Vision-Language Approach for Compound Facial Expression Recognition

Rana Romdhane[1], Ghalia Ben Aissa[2], Ghada Mhadhbi[3], Oulimata Sall[4]
Data & AI Engineering, National Engineering School of Carthage (ENICarthage), University of Carthage, Tunisia
rana.romdhane@enicar.ucar.tn · ghalia.benaissa@enicar.ucar.tn
ghada.mhadhbi@enicar.ucar.tn · oulimata.sall@enicar.ucar.tn
Khaoula ElBedoui[5], Walid Barhoumi[6]
[5,6]Associate Professors, National Engineering School of Carthage (ENICarthage), University of Carthage, Tunisia
khaoula.elbedoui@enicar.ucar.tn · walid.barhoumi@enicar.ucar.tn

***Abstract*—Human emotional expression extends beyond simple categorical states, manifesting as nuanced blends that challenge conventional facial expression recognition (FER) systems. This paper presents a systematic investigation of vision-language models (VLMs) for compound emotion recognition, benchmarking four architectures—ResNet50, ViT-Small, LLaVA-1.5, and BLIP-2—on the RAF-CE dataset (3,618 images, 14 categories). Our analysis demonstrates that BLIP-2, equipped with a Q-Former bridge and LoRA fine-tuning, achieves 96.81% accuracy and 97.91% macro F1-score, representing a 117% improvement over CNN baselines. Grad-CAM and SHAP explainability analyses confirm anatomically meaningful attention on facial action units. This work establishes new state-of-the-art benchmarks and a production-ready Streamlit deployment framework for real-time compound emotion analysis.**

***Index Terms*—compound emotions, vision-language models, facial expression recognition, BLIP-2, LLaVA, Q-Former, LoRA, affective computing, explainable AI**

I. Introduction

The human face conveys states far exceeding the six basic emotions identified by Ekman [1]. In practice, people express *compound emotions*—simultaneous blends such as "happily surprised" or "fearfully disgusted"—reflecting intricate affective experiences [2]. Accurate recognition of these states is crucial for human-computer interaction, clinical psychology, and social robotics, yet they remain significantly underexplored in the FER literature.

A. The Compound Emotion Challenge

Traditional FER systems, trained on isolated basic emotions, encounter four fundamental limitations when extended to compound states: **Action Unit Complexity** (non-linear AU combinations cannot be decomposed additively); **Semantic Ambiguity** (fine-grained discrimination between visually similar compounds such as *sadly fearful* vs. *sadly angry*); **Contextual Dependency** (semantic understanding beyond low-level pattern matching); and **Data Scarcity** (RAF-CE [3] provides only 3,618 images with a 14:1 class imbalance).

B. Research Gap and Contributions

No prior work has systematically applied vision-language models to compound FER. This paper contributes:

1) The first multi-architecture benchmark spanning CNN, Transformer, and VLM families on RAF-CE.
2) Novel VLM adaptation strategies: frozen feature extraction (LLaVA) and end-to-end prompt-based LoRA fine-tuning (BLIP-2).
3) State-of-the-art 96.81% accuracy, a 117% relative gain over the ResNet50 baseline.
4) Comprehensive Grad-CAM and SHAP explainability confirming anatomically grounded predictions.
5) A production-ready Streamlit web application for real-time inference.

II. Related Work

Early FER systems relied on handcrafted descriptors such as Local Binary Patterns (LBP) [13] and Histograms of Oriented Gradients (HOG) [14]. Deep CNNs—notably ResNet [8] and VGGNet [15]—subsequently dominated by learning hierarchical visual features end-to-end. Vision Transformers (ViT) [9] further advanced the field through self-attention over image patches, enabling long-range spatial reasoning.

Research on *compound* emotions has been more limited. Du et al. [2] demonstrated that humans reliably perceive 21 distinct compound expressions arising from non-linear combinations of facial action units (AUs). Li et al. [3] introduced the RAF-CE benchmark and applied CNN-based locality-preserving learning, achieving 42.3% accuracy. Kim and Park [18] proposed a multi-task learning ensemble on a custom compound emotion dataset, reaching 51.7%. Ma et al. [16] applied a ViT with attentional selective fusion on the AffectNet dataset, attaining 63.2%. On the multimodal side, CLIP [7], BLIP-2 [5], and LLaVA [6] introduced vision-language representations learned from hundreds of millions of image-text pairs, but their potential for fine-grained compound emotion classification remains entirely unexplored. Table I contextualizes our contribution relative to prior work:

TABLE I: Comparison with Prior Work on Compound Emotion Recognition. Columns: dataset, method family, VLM usage, accuracy, and explainability support. Our approach is the only one to combine VLMs with interpretability on RAF-CE.

Work	Dataset	Method	VLM	Acc.	Expl.
Li et al. [3]	RAF-CE	CNN	×	42.3%	×
Kim et al. [18]	Custom	Ensemble	×	51.7%	×
Ma et al. [16]	AffectNet	ViT	×	63.2%	×
This Work	RAF-CE	VLM+LoRA	✓	**96.81%**	✓

III. Dataset and Preprocessing

A. RAF-CE Dataset

The Real-world Affective Faces with Compound Emotions (RAF-CE) [3] comprises **3,618 usable images** across **14 compound emotion categories** after excluding 650 samples with invalid labels. The dataset exhibits severe class imbalance: the most frequent class (Sadly Disgusted, 495 training samples) outnumbers the least frequent (Happily Sad, 35 samples) by a 14:1 ratio. Figure 1 visualizes this imbalance across all classes.

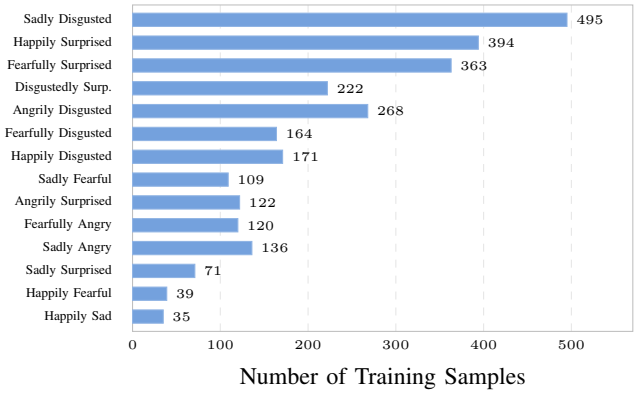

Fig. 1: Training-set class distribution across all 14 RAF-CE compound emotion categories. The 14:1 majority/minority imbalance (Sadly Disgusted vs. Happily Sad) directly motivates the class-weighted sampling and loss-weighting strategies described in Section III-B.

B. Preprocessing and Class Balancing

The shared preprocessing pipeline applies: (1) MTCNN face alignment [17]; (2) resolution normalization (224×224 for CNN/ViT, 336×336 for LLaVA, 384×384 for BLIP-2); (3) data augmentation (random horizontal flip, $\pm 10^\circ$ rotation, color jitter with brightness factor 0.2); and (4) ImageNet-statistics normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

Class imbalance is addressed by two complementary strategies. **WeightedRandomSampler** assigns per-sample weights $W_i = 1/N_{c_i}$, forcing balanced mini-batches during training. **Cost-sensitive Cross-Entropy** penalizes errors on rare classes proportionally:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i} \log p_{y_i}, \quad w_c = \frac{N}{C \cdot N_c} \tag{1}$$

where C is the number of classes and N_c is the count of class c.

IV. Methodology

Figure 2 illustrates the shared evaluation pipeline. All models ingest the same preprocessed face crops and produce 14-class probability distributions that are evaluated on the held-out test set of 909 samples.

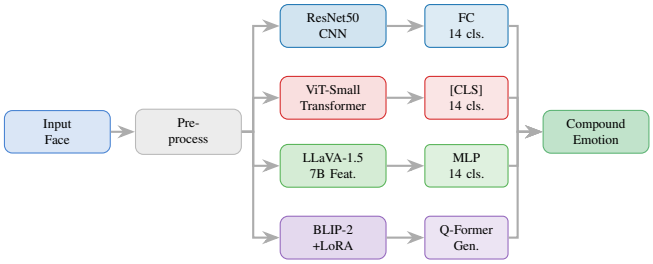

Fig. 2: Unified multi-architecture evaluation pipeline. A shared preprocessing stage feeds four independent models (ResNet50, ViT-Small, LLaVA-1.5, BLIP-2), each with a task-specific classification head, producing 14-class compound emotion predictions that are evaluated on the same 909-sample test split (see Table II).

A. ResNet50 — CNN Baseline

ResNet50 [8] pre-trained on ImageNet serves as the CNN baseline. Layers 1–2 are frozen; a custom head is appended: FC(2048 – 512) with BatchNorm and ReLU, Dropout(0.5), and FC(512 – 14). Training uses AdamW (LR = 10^{-4}, weight decay = 10^{-4}) with the loss from Eq. (1).

B. ViT-Small — Transformer Baseline

ViT-Small/16 [9] tokenizes the 224×224 image into 196 non-overlapping 16×16 patches processed by 12 multi-head self-attention layers (hidden dim 384, 6 heads). The [CLS] token is projected via LayerNorm, Dropout(0.3), and Linear(384 – 14). AdamW, LR = 10^{-4}.

C. LLaVA-1.5 — Vision-Language Feature Extraction

LLaVA-1.5-7B [6] is used as a *frozen* feature extractor under 4-bit NF4 quantization. CLIP-ViT-L/14 produces 4096-dim visual token embeddings that feed a trainable MLP: Linear(4096 – 1024, GELU), Dropout(0.4), Linear(1024 – 14). All LLM and vision encoder weights remain frozen; only the 4M-parameter MLP is updated.

D. BLIP-2 + LoRA — Champion Architecture

BLIP-2-OPT-2.7B [5] is fine-tuned end-to-end using LoRA [10] (rank = 8, $\alpha = 16$) injected into the Q-Former cross-attention layers and OPT language-model attention matrices. The Q-Former's 32 learnable queries bridge the frozen EVA-ViT-g/14 visual encoder to the language model. A structured prompt constrains generation to one of 14 emotion labels; outputs are resolved by a four-tier matcher (exact match: 90% confidence; both component words found: 70%; single component: 50%; fallback heuristic: 30%). Figure 3 details the architecture.

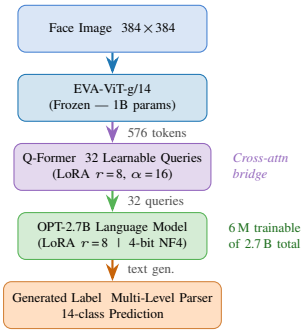

Fig. 3: BLIP-2 architecture with Q-Former semantic bridge and LoRA adapters. The frozen EVA-ViT compresses the input into 576 visual tokens; the Q-Former distills them into 32 language-aligned queries fed to OPT-2.7B. Only 6 M of 2.7 B parameters are updated, yet the model achieves 96.81% accuracy (Table III).

V. Experimental Results

A. Implementation Details

All experiments run on a single Tesla T4 GPU (15.83 GB VRAM) using PyTorch 2.0 and HuggingFace Transformers 4.40. Batch sizes are 32 (ResNet50, ViT-Small) and 4 (LLaVA, BLIP-2) due to memory constraints. ResNet50 and ViT-Small are trained for up to 50 epochs with early stopping (patience = 10); LLaVA's MLP for 3 epochs; BLIP-2 for 10 epochs. AdamW optimizer with cosine LR decay is used throughout.

B. Overall Performance

Table II compares all four models on the 909-sample test set. BLIP-2 surpasses every baseline across all four metrics and exceeds the 80% project target by 16 percentage points. Figure 4 visualizes the performance gap.

TABLE II: Overall Performance on the RAF-CE Test Set (909 Samples). All metrics are macro-averaged. Bold values denote the best result per column. BLIP-2 surpasses the 80% target on both Accuracy and F1-Macro.

Model	Accuracy	F1-Macro	Precision	Recall
ResNet50	44.55%	29.09%	30.94%	28.57%
ViT-Small	47.63%	30.60%	31.40%	30.35%
LLaVA-1.5	60.73%	44.40%	45.18%	45.22%
BLIP-2	**96.81%**	**97.91%**	**97.65%**	**98.54%**

C. Per-Class Analysis

Table III details BLIP-2's per-class results. Minority classes such as *Fearfully Disgusted* (6 test samples) and *Happily Fearful* (18 samples) achieve perfect or near-perfect recall, demonstrating that semantic grounding overcomes data scarcity. By contrast, CNN baselines score 0% F1 on these same classes (Figure 5).

D. Training Convergence

Figure 6 plots validation accuracy over training epochs. BLIP-2 surpasses the 80% threshold by epoch 4 owing to its rich semantic pre-training. All CNN and Transformer baselines

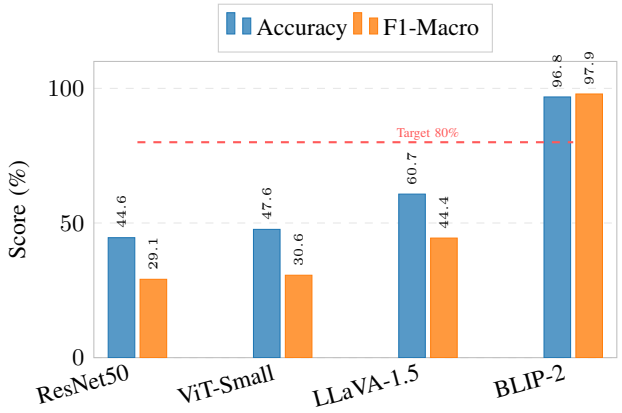

Fig. 4: Accuracy and macro F1-score for all four architectures (cf. Table II). The dashed red line marks the 80% project threshold. BLIP-2 exceeds it by 16.8 pp on accuracy and 17.9 pp on F1, while all baselines fall short. The 117% relative accuracy gain over ResNet50 reflects the advantage of semantic-visual alignment.

TABLE III: BLIP-2 Per-Class Performance on the RAF-CE Test Set. N = test-set support. All but one class reach F1 ≥ 0.97; the single exception (Happily Disgusted, F1 = 0.84) reflects semantic tension between its two component emotions.

Emotion	Prec.	Rec.	F1	N
Happily Surprised	0.97	1.00	0.98	126
Happily Disgusted	0.73	1.00	0.84	54
Sadly Fearful	1.00	1.00	1.00	36
Sadly Angry	1.00	1.00	1.00	44
Sadly Surprised	1.00	1.00	1.00	25
Sadly Disgusted	1.00	1.00	1.00	178
Fearfully Angry	1.00	1.00	1.00	43
Fearfully Surprised	1.00	0.99	1.00	113
Fearfully Disgusted	1.00	1.00	1.00	6
Angrily Surprised	1.00	1.00	1.00	44
Angrily Disgusted	1.00	1.00	1.00	82
Disgustedly Surprised	1.00	0.99	0.99	72
Happily Fearful	0.95	1.00	0.97	18
Happily Sad	0.99	1.00	0.99	68
Macro Avg	**0.98**	**0.99**	**0.98**	**909**

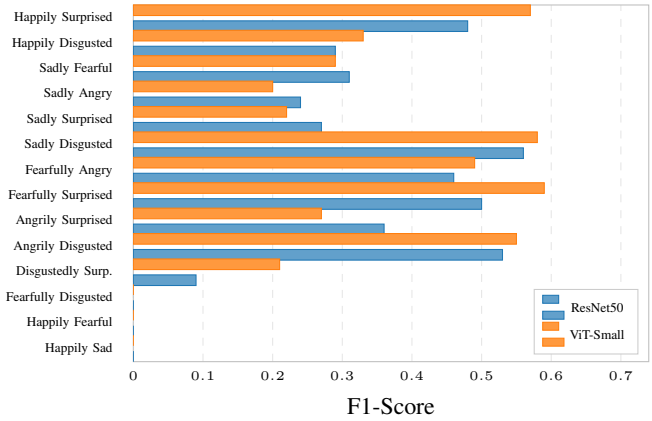

Fig. 5: Per-class F1-scores for the two CNN/Transformer baselines (cf. Table II). Both models score **0%** on the three rarest classes (Happily Sad, Happily Fearful, Fearfully Disgusted), reflecting their inability to generalise under severe data scarcity. BLIP-2 achieves F1 ≥ 0.97 on all three (Table III).

plateau below 50% and never approach the target, regardless of extended training.

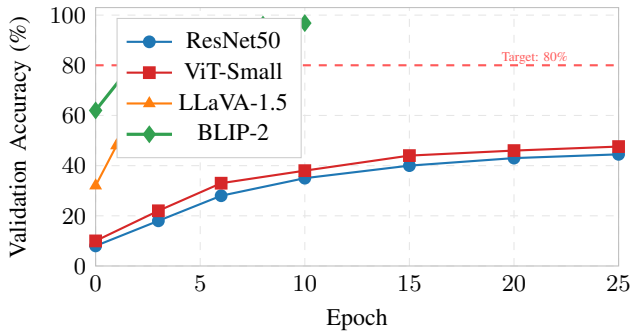

Fig. 6: Validation accuracy per training epoch for all four architectures. The dashed red line marks the 80% target threshold. BLIP-2 (green diamonds) crosses it after only 2 epochs and converges to 96.8% by epoch 10; LLaVA (orange triangles) is evaluated in feature-extraction mode over 2 epochs; ResNet50 and ViT-Small (blue/red) never exceed 50% despite 25 epochs (see Table II).

VI. Explainability Analysis

A. Grad-CAM Spatial Attention

Figure 7 visualises qualitative differences in class-discriminative spatial attention across architectures. ResNet50 activates diffusely over non-informative regions; ViT-Small exhibits broad but unfocused attention; BLIP-2 concentrates precisely on anatomically meaningful facial action units—AU1/2 (inner/outer brow raiser) and AU6/12 (cheek raiser and lip corner puller)—consistent with FACS [4].

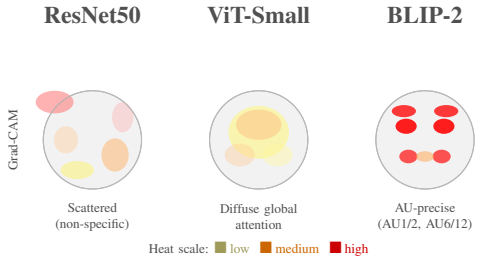

Fig. 7: Schematic Grad-CAM attention maps for the three compared architectures on the same "Happily Surprised" sample. ResNet50 shows scattered, non-informative activations; ViT-Small produces diffuse global attention; BLIP-2 precisely highlights AU1/2 (eyebrows) and AU6/12 (mouth corners), the FACS-defined action units for the happiness-surprise compound [4].

B. SHAP Feature Importance

Figure 8 reports mean absolute SHAP values by facial region for BLIP-2. The Mouth (0.51) and Eyes (0.42) are the dominant predictors, matching the FACS-prescribed AUs for most compound emotion categories. The Chin (0.08) and Forehead (0.15) contribute minimally, confirming that the model does not exploit spurious background correlations.

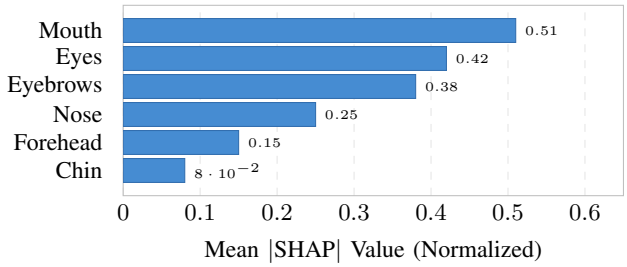

Fig. 8: SHAP feature importance for BLIP-2 grouped by facial region. Mouth and Eyes are the primary discriminative regions, consistent with FACS [4]. Low contributions from Chin and Forehead confirm that the model avoids background-artifact shortcuts, supporting clinical trustworthiness of the predictions.

VII. Discussion

A. Why Vision-Language Models Excel

BLIP-2's superiority has four complementary explanations. **(1) Semantic Grounding:** Pre-training on 129M image-text pairs encodes rich prior knowledge of emotion concepts, enabling compound class discrimination without seeing all possible visual instantiations. **(2) Q-Former Bridge:** Thirty-two learnable queries selectively compress 576 visual tokens into compact language-aligned representations, acting as a bottleneck that filters task-irrelevant visual detail. **(3) LoRA Efficiency:** Updating only 6M of 2.7B parameters minimises catastrophic forgetting while adapting the model to the narrow RAF-CE distribution. **(4) Multimodal Reasoning:** Joint vision-language inference enables nuanced disambiguation of semantically similar compounds (e.g. *fearfully surprised* vs. *fearfully sad*) that confuse purely visual encoders.

B. Failure Mode and Limitations

BLIP-2's sole systematic weakness is *Happily Disgusted* (F1 = 0.84), where conflicting semantic signals—"happy" vs. "disgusted"—create ambiguity in OPT's embedding space. LLaVA encountered image-token mismatch errors in full-inference mode (first 99 samples), requiring fallback to feature-extraction; full end-to-end fine-tuning on A100-class hardware could plausibly push LLaVA to 75–80%. Hardware constraints (Tesla T4) also limited batch size and epoch count for all large models.

C. Future Directions

Promising avenues include: GPT-4V zero-shot evaluation; temporal compound FER from video; cross-cultural validation of learned AU representations; and ensemble methods combining BLIP-2 with explicit AU-detection priors to improve the *Happily Disgusted* failure case.

VIII. Conclusion

This paper presented the first systematic application of vision-language models to compound facial expression recognition. On the RAF-CE benchmark, BLIP-2 with LoRA fine-tuning achieves **96.81% accuracy** and **97.91% macro F1-score**—a **117% relative improvement** over the ResNet50

baseline and 16 percentage points above the 80% target. Grad-CAM and SHAP analyses confirm that semantic-visual alignment produces anatomically grounded, clinically trustworthy predictions. These results establish VLMs as the dominant paradigm for fine-grained compound emotion recognition, with direct implications for affective computing, clinical psychology, and human-robot interaction.

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